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"HIDDEN MARKOV MODEL"

 $\lambda = 1, 6, 6, 5, 6, 2, 6, 6, 3, 6$ $\pi = F F L L F F F L L L$

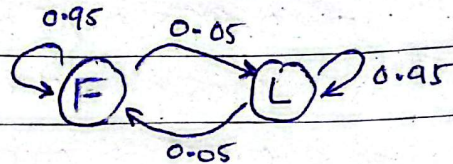
→ Probability of 1 given that
it is Fair Dice

$$P(1|F) = 1/6$$

$$P(2|F) = 1/6$$

$$P(5|L) = 1/10$$

$$P(6|L) = 1/2$$



$$P(\lambda, \pi) = \text{Initial Prob} \times \text{Emission Prob} \times \text{Transition Prob}$$

↳ usually given

if not assume 0.5

Emission Probability:

$$E(\lambda, \pi) = P(1|F) \times P(6|F) \times P(6|L) \times P(5|L) \times$$

$$P(6|F) \times P(2|F) \times P(6|F) \times P(6|L) \times$$

$$P(3|L) \times P(6|L)$$

$$= 1/6 \times 1/6 \times 1/2 \times 1/10 \times 1/6 \times 1/6 \times 1/6 \times$$

$$1/2 \times 1/10 \times 1/2$$

$$= (1/6)^5 \times (1/2)^3 \times (1/10)^2$$

$$= 1.607 \times 10^{-7}$$

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Transition Probability :

→ Probability the dice is loaded given that the previous die was also loaded.

$$= P(R|R) \times P(L|R) \times P(L|L) \times P(R|L) \times P(R|R) \times$$
$$P(R|R) \times P(L|R) \times P(L|L) \times P(L|L)$$

$$= 0.95 \times 0.05 \times 0.95 \times 0.05 \times 0.95 \times 0.95 \times$$
$$0.05 \times 0.95 \times 0.95$$

$$= 0.95^6 \times 0.05^3$$

$$= 9.188 \times 10^{-5}$$

$$P(x, \pi) = 0.5 \times 1.607 \times 10^{-7} \times 9.188 \times 10^{-5}$$

$$\boxed{P(x, \pi) = 7.382 \times 10^{-12}}$$